

Lifestyle & Health: Disease Risk

Quick reference sheet · Biology · keep handy for revision

1. Communicable vs non-communicable disease

A **communicable** disease is caused by a pathogen and can spread between people. A **non-communicable** disease cannot spread between people — it develops from genetic, lifestyle and/or environmental factors, often over a long time (e.g. cardiovascular disease, type 2 diabetes, some cancers, chronic lung disease).

2. Risk factors for non-communicable disease

A **risk factor** increases the probability of developing a disease — it does not guarantee it.

Risk factor	Linked to
Smoking	Lung disease, lung cancer, cardiovascular disease
Poor diet (fat/salt/sugar)	Obesity, type 2 diabetes, cardiovascular disease
Excess alcohol	Liver disease, some cancers
Lack of exercise	Obesity, cardiovascular disease, type 2 diabetes
Obesity	Type 2 diabetes, cardiovascular disease, some cancers

Correlation is not cause: a link between a factor and a disease doesn't prove the factor causes it — look for a plausible mechanism and repeated studies.

3. Cardiovascular disease & treatments

In coronary heart disease, fatty deposits (atheroma) narrow the coronary arteries, restricting blood flow to the heart muscle.

Treatment	What it does
Statins	Drugs that reduce blood cholesterol, slowing fatty deposit build-up
Stents	Mesh tubes that hold a narrowed artery open
Bypass surgery	A healthy vessel bypasses the blocked section of a coronary artery
Artificial hearts/valves	Mechanical devices that pump blood or replace a faulty valve
Heart transplant	A diseased heart is replaced with a healthy donor heart

Evaluate: drugs/stents are less invasive but ongoing or can narrow again; surgery/transplant is more direct but carries surgical risk and donor hearts are scarce.

4. Homeostasis — the big idea

Homeostasis is the maintenance of a constant internal environment (e.g. body temperature, blood glucose, water content). Pattern: receptor detects a change → coordination system (nerves/hormones) → effector responds to restore the normal level. Detailed hormone control (e.g. insulin) is covered in Part 2.